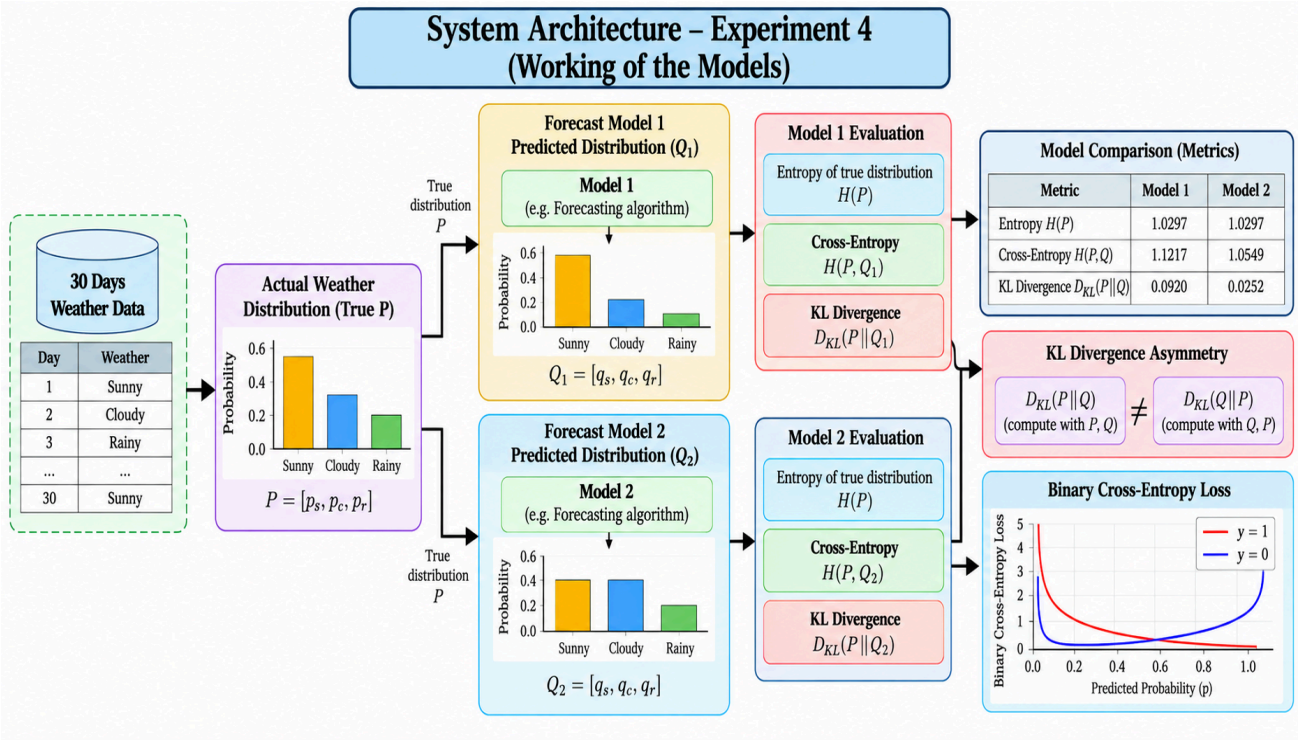
Experiment 4: Entropy, Cross-Entropy and KL Divergence

1. AIM

To calculate entropy, cross-entropy, and KL divergence for observed weather distributions and two forecasting models. To verify the relationship between cross-entropy, entropy, and KL divergence, demonstrate the asymmetry of KL divergence, and plot binary cross-entropy loss against predicted probability using NumPy, SciPy, and Matplotlib.

SYSTEM ARCHITECTURE



2. MATHEMATICAL FORMULAS

1. Entropy

Entropy measures the uncertainty in the actual probability distribution.

$$\boxed{H(P) = -\sum_{i=1}^{n} P(y_i) \log P(y_i)}$$

2. Cross-Entropy

Cross-entropy measures how well a predicted distribution Q represents the actual distribution P.

$$\boxed{H(P, Q) = -\sum_{i=1}^{n} P(y_i) \log Q(y_i)}$$

3. KL Divergence

KL divergence measures the difference between the actual and predicted probability distributions.

$$\boxed{D_{KL}(P \| Q) = \sum_{i=1}^{n} P(y_i) \log \frac{P(y_i)}{Q(y_i)}}$$

4. Relationship Between Cross-Entropy, Entropy and KL Divergence

$$\boxed{H(P, Q) = H(P) + D_{KL}(P \| Q)}$$

5. KL Divergence Asymmetry

$$\boxed{D_{KL}(P \| Q) \neq D_{KL}(Q \| P)}$$

6. Binary Cross-Entropy Loss

$$\boxed{L = -[y \log(p) + (1 - y) \log(1 - p)]}$$

CODE

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import entropy

# Observed weather distribution: Sunny, Cloudy, Rainy
P = np.array([0.5, 0.3, 0.2])

# Two forecasting models
Q1 = np.array([0.7, 0.2, 0.1])
Q2 = np.array([0.4, 0.4, 0.2])

# 1. Entropy of observed distribution
H_P = entropy(P)

print("Entropy of observed distribution:", round(H_P, 4))

# 2. Cross-entropy and KL divergence
def cross_entropy(P, Q):
    return -np.sum(P * np.log(Q))

CE1 = cross_entropy(P, Q1)
CE2 = cross_entropy(P, Q2)

KL1 = entropy(P, Q1)
KL2 = entropy(P, Q2)

print("\nModel 1")
print("Cross-Entropy:", round(CE1, 4))
print("KL Divergence:", round(KL1, 4))

print("\nModel 2")
print("Cross-Entropy:", round(CE2, 4))
print("KL Divergence:", round(KL2, 4))

# 3. Verify cross-entropy = entropy + KL divergence
print("\nVerification")
print("Model 1:", np.isclose(CE1, H_P + KL1))
print("Model 2:", np.isclose(CE2, H_P + KL2))

# 4. Verify KL divergence asymmetry
KL_PQ = entropy(P, Q1)
```

```
KL_QP = entropy(Q1, P)

print("\nKL Divergence Asymmetry")
print("KL(P || Q1):", round(KL_PQ, 4))
print("KL(Q1 || P):", round(KL_QP, 4))

# 5. Plot probability distributions
labels = ["Sunny", "Cloudy", "Rainy"]
x = np.arange(len(labels))
width = 0.25

plt.figure(figsize=(8, 5))
plt.bar(x - width, P, width, label="Observed P")
plt.bar(x, Q1, width, label="Model 1")
plt.bar(x + width, Q2, width, label="Model 2")

plt.xticks(x, labels)
plt.ylabel("Probability")
plt.title("Observed vs Predicted Weather Distributions")
plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.show()

# 6. Binary cross-entropy loss
p = np.linspace(0.001, 0.999, 500)

loss_y1 = -np.log(p)
loss_y0 = -np.log(1 - p)

plt.figure(figsize=(8, 5))
plt.plot(p, loss_y1, label="Actual class y = 1")
plt.plot(p, loss_y0, label="Actual class y = 0")

plt.xlabel("Predicted Probability")
plt.ylabel("Binary Cross-Entropy Loss")
plt.title("Binary Cross-Entropy Loss vs Predicted Probability")
plt.legend()
plt.grid(True, alpha=0.3)
plt.ylim(0, 5)
plt.show()
```

OUTPUT

Entropy of Observed Distribution

Given:

$P = [0.5, 0.3, 0.2]$
$H(P) = -[0.5\ln(0.5) + 0.3\ln(0.3) + 0.2\ln(0.2)]$
H(P)=1.0297

Cross-Entropy and KL Divergence

Metric	**Model 1**	**Model 2**
Entropy $H(P)$	1.0297	1.0297
Cross-Entropy $H(P, Q)$	1.1217	1.0549
KL Divergence $D_{KL}(P\|Q)$	0.0920	0.0252

For Model 1:

$$H(P) + D_{KL}(P\|Q_1) = 1.0297 + 0.0920$$

$$\boxed{H(P, Q_1) = 1.1217}$$

For Model 2:

$$H(P) + D_{KL}(P\|Q_2) = 1.0297 + 0.0252$$

$$\boxed{H(P, Q_2) = 1.0549}$$

Both satisfy the relationship:

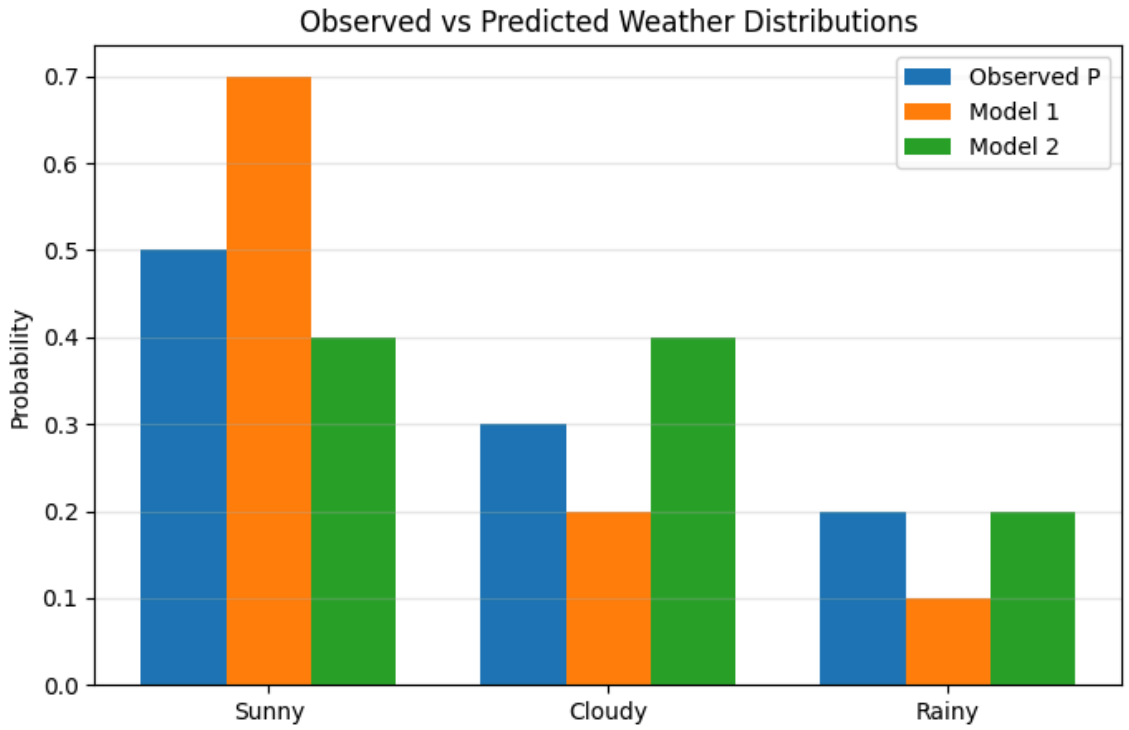
$$\boxed{H(P, Q) = H(P) + D_{KL}(P\|Q)}$$

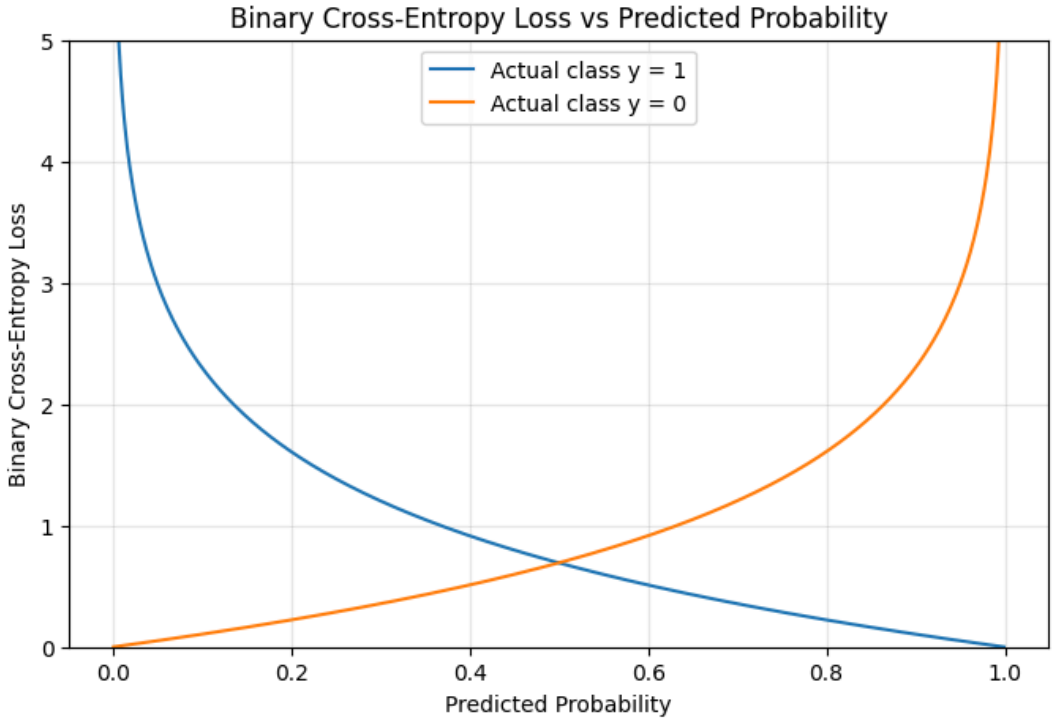
D. KL Divergence Asymmetry

For Model 1:

$$D_{KL}(P\|Q_1) \approx 0.0920$$

$$D_{KL}(Q_1\|P) \approx 0.0851$$





RESULT

Entropy, cross-entropy, and KL divergence were calculated for the observed weather distribution and two forecasting models. The relationship (H(P,Q)=H(P)+D_{KL}(P\|Q)) was verified for both models.